UNIVERSITY OF TEXAS ARLINGTON
COLLEGE OF SCIENCE

Advanced Classical Mechanics

Lecture 5: Newtonian Dynamics

Readings: Morin - Chapter 3, Marion Chapter 2

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Newtonian Dynamics (by Isaac Newton at age 44 in 1687)

Inertial Frame, defined by Newton's first law of motion (and Galilean transformation), is a reference frame within which Newton's (first and second) laws of motion hold.

Newton's third law is a manifestation of the law of conservation of momentum:

$$\frac{d\boldsymbol{P}}{dt} = \frac{d\boldsymbol{P_1}}{dt} + \frac{d\boldsymbol{P_2}}{dt} = \boldsymbol{F}_1 + \boldsymbol{F}_2 = 0 \;\Rightarrow \boldsymbol{F}_1 = -\boldsymbol{F}_2$$

Intuitively, the third law can be also expressed as the phenomenon that we will never find a particle accelerating unless there's some other particle accelerating somewhere else.

Problem solving strategies in Newtonian Dynamics:

- Draw free-body diagrams: The term free-body diagram is used to denote a diagram with all the forces drawn on a given object. After drawing such a diagram for each object in the setup, we simply write down all the $\boldsymbol{F} = m\boldsymbol{a}$ equations they imply.

- Solve the differential equation $\boldsymbol{F}(\boldsymbol{v}, \boldsymbol{x}, t) = m\boldsymbol{a} = \frac{\boldsymbol{d^2}\ddot{\boldsymbol{x}}}{\boldsymbol{dt^2}}$

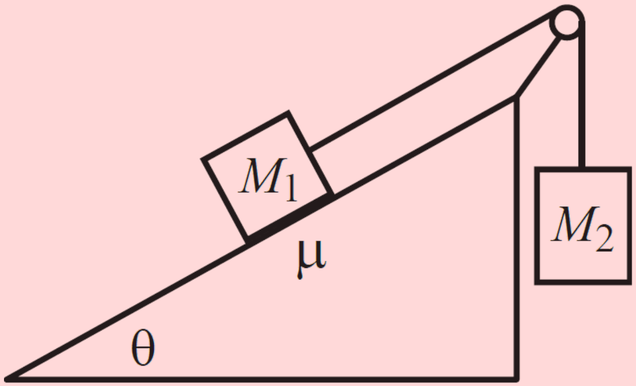
To solve the above differential equation, three knowns must be always given in the problem: Three initial conditions:

  - t_0: An initial reference time
  - $\boldsymbol{x}_0 \equiv \boldsymbol{x}(t_0)$: the initial location of the system experiencing $\boldsymbol{F}(\boldsymbol{v}, \boldsymbol{x}, t)$ at time t_0
  - $\boldsymbol{v}_0 \equiv \boldsymbol{v}(t_0)$: the initial velocity of the system experiencing $\boldsymbol{F}(\boldsymbol{v}, \boldsymbol{x}, t)$ at time t_0

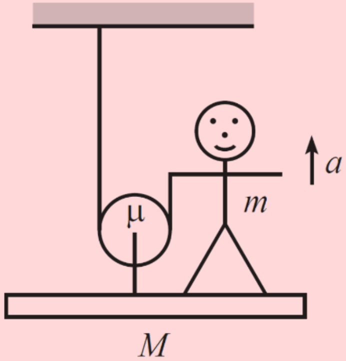
If we are lucky, the differential equation can be solved analytically, otherwise, it must be solved numerically using a computer.

Newtonian Dynamics

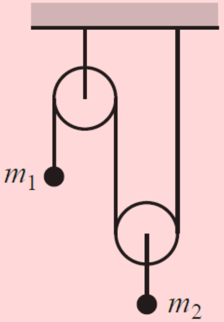
Problem 1. (A plane and masses): Mass M_1 is held on a plane with Inclination angle θ, and mass M_2 hangs over the side. The two masses are connected by a massless string which runs over a massless pulley (see the Figure). The coefficient of kinetic friction between M_1 and the plane is μ. M_1 is released from rest. Assuming that M_2 is sufficiently large so that M_1 gets pulled up the plane, what is the acceleration of the masses? What is the tension in the string?



Problem 2. (Platform and pulley): A person stands on a platform and pulley system, as shown in the figure. The platform, person, and pulley masses are M, m, and μ, respectively. The rope is massless. Let the person pull up on the rope so that she has acceleration a upward. (Assume that the platform is somehow constrained to stay level, perhaps by having the ends run along some rails.) Find the tension in the rope, the normal force between the person and the platform, and the tension in the rod connecting the pulley to the platform (ignore rotational dynamics).



Problem 3. (Atwood's machine): Consider the pulley system in the figure, with masses m_1 and m_2. The strings and pulleys are massless. What are the accelerations of the masses? What is the tension in the string?



Newtonian Dynamics

Problem 4. (Differential Equation: Gravitational force):
A particle of mass m is subject to a constant force $F = -mg$.
The particle starts at rest at height h. Because this constant force falls into all of the above three categories, we should be able to solve for y(t) in two ways:
(a) Find $y(t)$ by writing a as dv/dt.
(b) Find $y(t)$ by writing a as vdv/dy.

Problem 5. (Differential Equation: Dropped ball): A beach ball is dropped from rest at height h. The air drag force is $F_d = -\beta v$. Find the velocity and height as a function of time.

Problem 6. (Differential Equation: Projectile motion): Consider a ball thrown through the air, not necessarily vertically. We will neglect air resistance in the following discussion. Let x and y be the horizontal and vertical positions, respectively. Assuming the initial position and velocity are (X, Y) and (V_x, V_y), derive the equations of motion of this projectile.

Problem 7. (Throwing a ball): (a) For a given initial speed, at what inclination angle should a ball be thrown so that it travels the maximum horizontal distance by the time it returns to the ground? Assume that the ground is horizontal, and that the ball is released from ground level. **(b)** What is the optimal angle if the ground is sloped upward at an angle β (or downward, if β is negative)?

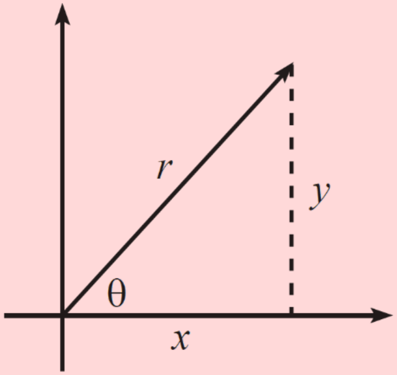
Problem 8. (Differential Equation: Projectile with drag): A ball is thrown with speed v_0 at an angle θ. Let the drag force from the air take the form $F_d = -\beta v \equiv -m\alpha v$.
(a) Find $x(t)$ and $y(t)$.
(b) Assume that the drag coefficient takes the value that makes the magnitude of the initial drag force equal to the weight of the ball. If your goal is to have x be as large as possible when y achieves its maximum value (you don't care what this maximum value actually is), show that θ should satisfy $\sin\theta = (\sqrt{5} - 1)/2$, which just happens to be the inverse of the golden ratio.

Newtonian Dynamics

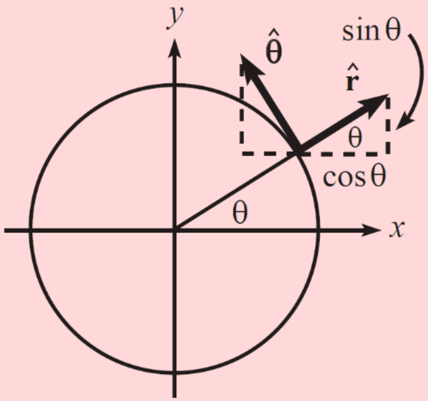
Problem 9. (Force components in polar coordinates):
Recall the definition of Polar Coordinates from Question 6 of our lecture 1 (ACM1). The Cartesian Coordinates can be written in terms Of polar coordinates as,

$$x = r\cos\theta \;;\, y = r\sin\theta \;.$$

We would like to express the equation of motion in Polar Coordinates as opposed to Cartesian Coordinates ($\boldsymbol{F} = m\boldsymbol{a} \equiv m\ddot{\boldsymbol{r}}$).



Using the representation of the Polar bases as illustrated in the figure, show that the Polar equation of motion can be written as,

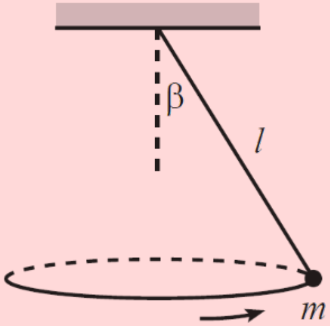


$$F \equiv F_r\hat{\boldsymbol{r}} + F_\theta\widehat{\boldsymbol{\theta}} = m\ddot{\boldsymbol{r}},$$

where,

$$F_r = m\left(\ddot{r} - r\dot{\theta}^2\right),$$
$$F_\theta = m\left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right).$$

Problem 10. (Circular pendulum): A mass hangs from a massless string of length l. Conditions have been set up so that the mass swings around in a horizontal circle, with the string making a constant angle β with the vertical. What is the angular frequency, ω, of this motion?



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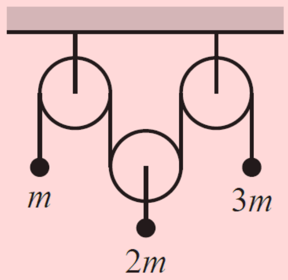
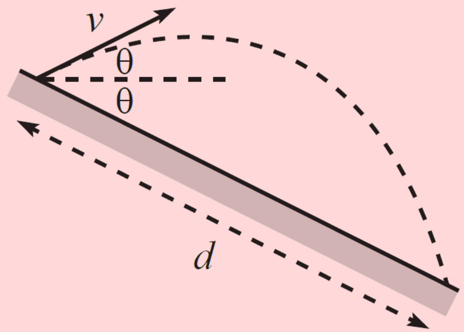
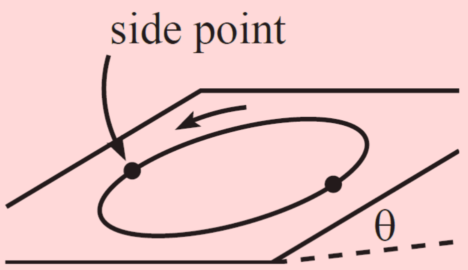
Problem 11. (Atwood's Machine):
Consider the Atwood's machine in the figure, with masses m, $2m$, and $3m$.
Find the accelerations of the masses.



Problem 12. (Equal tilts):
A plane tilts down at an angle θ below the horizontal. On this plane, a projectile is fired with speed v at an angle θ above the horizontal, as shown in the figure. What is the distance, d, along the plane that the projectile travels? What is d in the limit $\theta \to 90^\circ$? What θ yields the maximum horizontal distance?



Problem 13. (Driving on tilted ground):
A plane tilts down at an angle θ below the horizontal. On this plane, a projectile is fired with speed v at an angle θ above the horizontal, as shown in the figure. What is the distance, d, along the plane that the projectile travels? What is d in the limit $\theta \to 90^\circ$? What θ yields the maximum horizontal distance?



Problem 14. (A force $\boldsymbol{F_\theta = 2m\dot{r}\dot{\theta}}$): Consider a particle that feels an angular force only, of the form $F_\theta = 2m\dot{r}\dot{\theta}$. Show that $r = Ae^{\theta} + B^{-\theta}$, where A and B are constants of integration, determined by the initial conditions.